

ENV S 193DS Final

Stella Symonds

2026-06-09

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https://github.com/stellassymonds/ENVS-193DS_spring-2026_final

Setup

```
# read in packages
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(DHARMA) # for residual plots
library(ggeffects) # getting model predictions
library(MuMIn) # model selection
library(effsize) # effect size

# read in data
nests <- read.csv(here("data", "nest_data_final.csv"))
bike_rides <- read.csv(here("data", "bike_rides.csv"))
```

Problems

Problem 1

- a. In part 1, they used a generalized linear model. In part 2, they used a chi-squared test.
- b. Part 1: We found evidence to suggest that the distance to water edge does predict probability of American Avocet microhabitat use.

Part 2: We found evidence to suggest that there is a difference between American Avocet, Forster's Tern, and Black-necked Stilt bird species' habitat preferences of managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat habitats.

- c. Associated behavior type of either feeding, breeding, or roosting could be a categorical variable that influences the probability of American Avocet microhabitat use. For example, waterbirds that are breeding or roosting may prefer islands within ponds.

Problem 2

- a.

```
# create new object from nests
nests_clean <- nests |>
  # reordering ant species
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "AB", # C. sjostedti
                                   "RRB", # C. mimosae
                                   "BBR")) |> # C. nigriceps

# select for preferred columns
select(height_cm, ant_species, case_control) |>
# filter for preferred ant species
filter(ant_species %in% c("RRB", "BBR", "AB")) |>
# create new column with full scientific name
mutate(scientific_name = case_when(
  ant_species == "AB" ~ "C. sjostedti",
  ant_species == "RRB" ~ "C. mimosae",
  ant_species == "BBR" ~ "C. nigriceps"))

slice_sample(nests_clean, # data frame to display
             n = 10) # select ten random rows
```

	height_cm	ant_species	case_control	scientific_name
1	359.5	RRB	0	C. mimosae
2	410.0	RRB	0	C. mimosae
3	166.0	RRB	1	C. mimosae
4	389.0	RRB	1	C. mimosae
5	248.5	RRB	1	C. mimosae
6	167.0	BBR	0	C. nigriceps
7	117.5	RRB	0	C. mimosae
8	423.0	RRB	0	C. mimosae
9	157.0	RRB	0	C. mimosae
10	427.0	RRB	0	C. mimosae

b. 1s mean that the tree contained a nest and 0s mean the tree did not contain a nest and was available for comparison.

c. Tree height is a continuous variable. As tree height increases, the probability of nest occurrence would increase because they are farther from possible nest predators or disturbances (Results, paragraph 2).

Acacia ant species is a categorical variable. If an ant species is a more aggressive defender, the probability of nest occurrence would increase because the ant species reduces the risk of nest predation (Introduction, paragraph 3). If acacia ants defend against all possible tree disturbances, then the probability of nest occurrence in trees with less aggressive ant species would increase (Introduction, paragraph 3).

d. 2 models total:

Model number	Tree height	Ant Species	Interaction?
1	X	X	no
2	X	X	yes

e.

f.

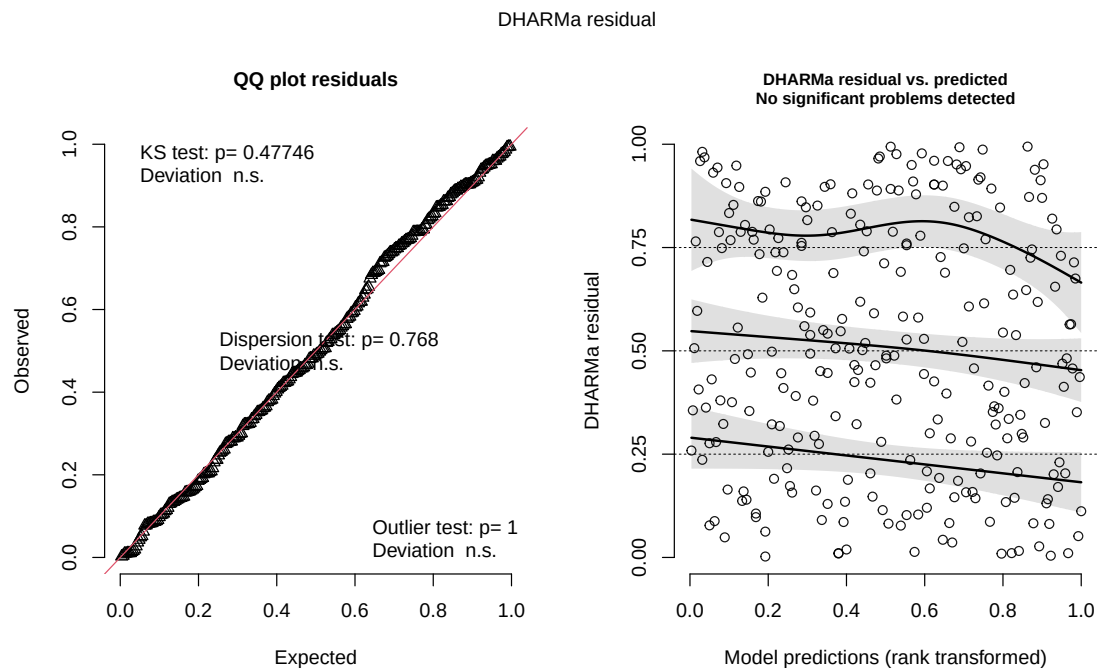
```
AICc( # calculating AIC of both models
  model1,
  model2) |>
# arranging output in order of AIC
arrange(AICc)
```

	df	AICc
model2	6	238.1236
model1	4	258.8940

The best model as determined by Akaike's Information Criterion (AIC) shows nest occurrence as a function of tree height in cm interacting with ant species.

g.

```
# generating diagnostic plots for best model (by AIC value) from DHARMa
plot(
  simulateResiduals(model2)
)
```



h.

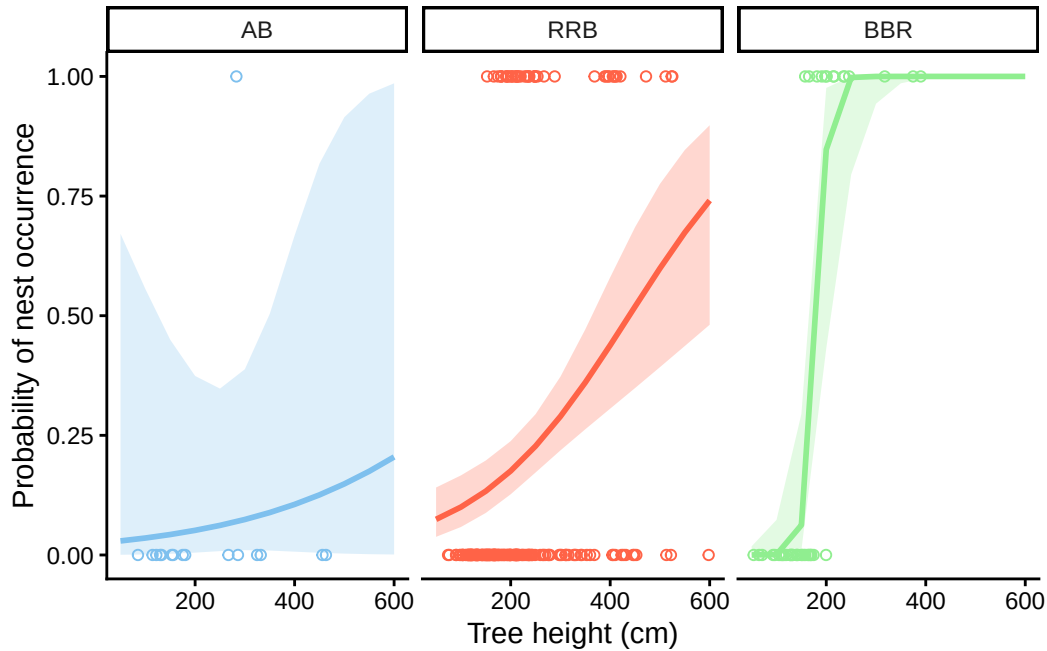
```
# create prediction
preds <- ggpredict(
  model2, # specify model
  terms = c("height_cm", "ant_species")) |>
  # renaming columns
  rename(height_cm = x,
          ant_species = group)

# creating visualization
ggplot(data = nests_clean, # nests_clean data frame
```

```

    mapping = aes(x = height_cm, # x is tree height
                  y = case_control, # y is nest occurrence
                  color = ant_species)) + # color by species
# create separate panels for each species
facet_wrap(~ ant_species) +
# first layer: jitterplot to show data
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) + # open circles
# second layer: ribbons for confidence intervals
geom_ribbon(data = preds, # specify prediction data frame
           mapping = aes(x = height_cm, # x axis is height
                         ymin = conf.low, # lower confidence bound
                         ymax = conf.high, # higher confidence bound
                         fill = ant_species), # do by species
           alpha = 0.25, # semi-transparent so raw data shows through
           inherit.aes = FALSE) + # don't call from preds aes
# third layer: line
geom_line(data = preds, # specify prediction data frame
         mapping = aes(x = height_cm, # x axis is height
                       y = predicted, # y s predicted values
                       color = ant_species), # color by ant species
         linewidth = 1,
         inherit.aes = FALSE) + # don't call from preds aes
# rename axes with units
labs(x = "Tree height (cm)",
     y = "Probability of nest occurrence") +
theme(panel.grid = element_blank()) + # take out gridlines
theme_classic() + # bye gray background
theme(legend.position = "none") + # taking out legend
# custom colors for lines and confidence intervals
scale_color_manual(values = c("RRB" = "tomato",
                              "AB" = "skyblue2",
                              "BBR" = "lightgreen")) +
scale_fill_manual(values = c("RRB" = "tomato",
                              "AB" = "skyblue2",
                              "BBR" = "lightgreen"))

```



i. Figure 1. Nest occurrence as a function of tree height in cm interacting with ant species. Open circles represent individual observations. Lines represent model predictions. Transparent ribbons represent a 95% confidence interval. . Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

j.

```
# calculate the probability of nest occurrence at height = 600 cm
ggpredict(model2,           # model object
  terms = c("height_cm[600]", "ant_species")) # "predictor" and predictor value in brackets

# Predicted probabilities of case_control

ant_species: AB

height_cm | Predicted |    95% CI
-----|-----|-----
600 |      0.20 | 0.00, 0.99

ant_species: RRB
```

height_cm	Predicted	95% CI
600	0.74	0.48, 0.90

ant_species: BBR

height_cm	Predicted	95% CI
600	1.00	1.00, 1.00

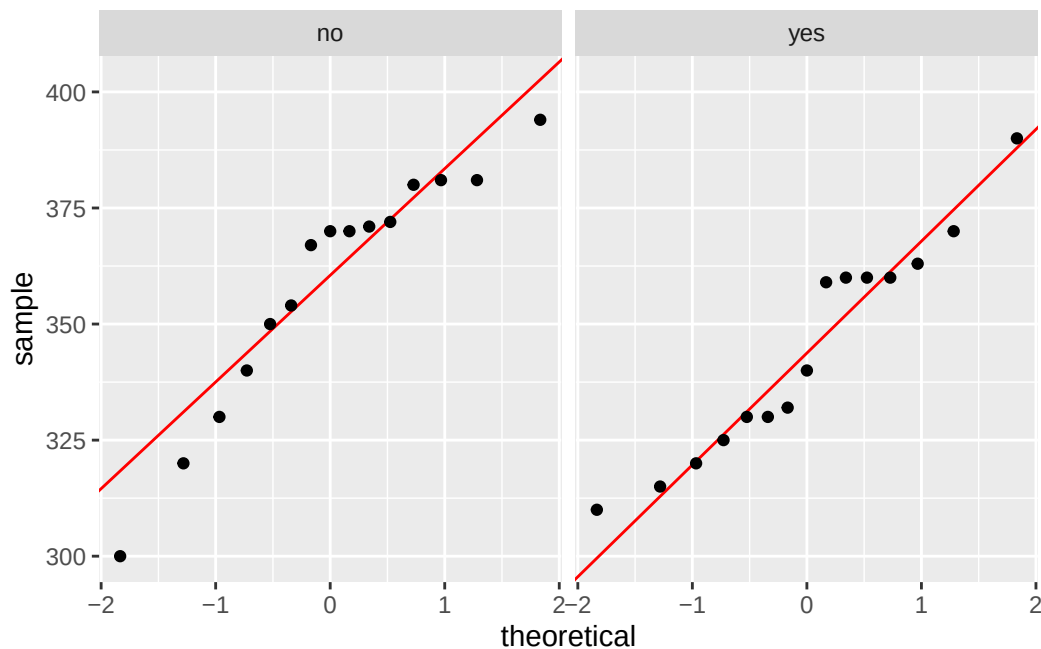
k.

Problem 3

- My affective visualization would be impossible to analyze without a key accompanying it. I prioritized creativity over feasibility of understanding. My exploratory visualizations are very easy to understand and make conclusions (or lack thereof) from. All my visualizations showed close to an equal amount of bike rides when I did and didn't have plans after class and low, medium, or high wind speed. The variation in data was consistent and can be seen in all visualizations. A slight difference in medians between bike rides when I have plans after class and no plans after class is visible in my exploratory visualization, but is not clear in my affective visualization. This is because my data is oriented around a central point in my affective visualization. I don't think a clear association between wind speed and bike ride duration is obvious in any of my visualizations. My feedback in workshop consisted of possibly adding icons to the end of the lines to represent temperature or another variable. I ultimately decided to not implement this feedback, since neither temperature nor other variables I tracked had any visible affect on my data, and I thought it may make it harder to see the difference between the lengths of each stitched line.
- A two sample t-test is appropriate to test whether there is a difference in mean bike ride duration between two groups, categorized by status of my plans after class (yes/no). My plans status after class is a binary variable, meaning it can be split into two groups, and my bike ride duration is a continuous variable.
-

```
# cleaning data
bike_rides_clean <- bike_rides |> # create new object from rides
clean_names() |> # clean column names
# select plans and duration columns
select(plans_after_class, duration_seconds)
```

```
# normality check
# base layer: ggplot
ggplot(data = bike_rides_clean, # starting data frame
       aes(sample = duration_seconds)) + # y-axis for QQ plot
  # first layer: QQ reference line
  geom_qq_line(color = "red") + # adding a color
  geom_qq() + # second layer: QQ plot
  # creating "facets"
  facet_wrap(~ plans_after_class) # show treatments separately
```



```
# check variances with F test of equal variances
var.test(
  duration_seconds ~ plans_after_class, # formula: response variable ~ grouping variable
  data = bike_rides_clean) # bike_rides_clean data frame
```

F test to compare two variances

```
data: duration_seconds by plans_after_class
F = 1.253, num df = 14, denom df = 14, p-value = 0.6789
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
```



```
0.4206592 3.7320819
sample estimates:
ratio of variances
      1.25297
```

```
t.test(
  # formula: response variable ~ grouping variable
  duration_seconds ~ plans_after_class,
  var.equal = TRUE, # argument for unequal variances
  data = bike_rides_clean) # bike_rides_clean data frame
```

Two Sample t-test

```
data: duration_seconds by plans_after_class
t = 1.5915, df = 28, p-value = 0.1227
alternative hypothesis: true difference in means between group no and group yes is not equal
95 percent confidence interval:
 -4.134173 32.934173
sample estimates:
mean in group no mean in group yes
      358.6667      344.2667
```

```
# calculate cohen's d
d <- cohen.d(duration_seconds ~ plans_after_class, data = bike_rides_clean)

d # display effect size value
```

Cohen's d

```
d estimate: 0.5811322 (medium)
95 percent confidence interval:
      lower      upper
-0.1824647  1.3447291
```

d.

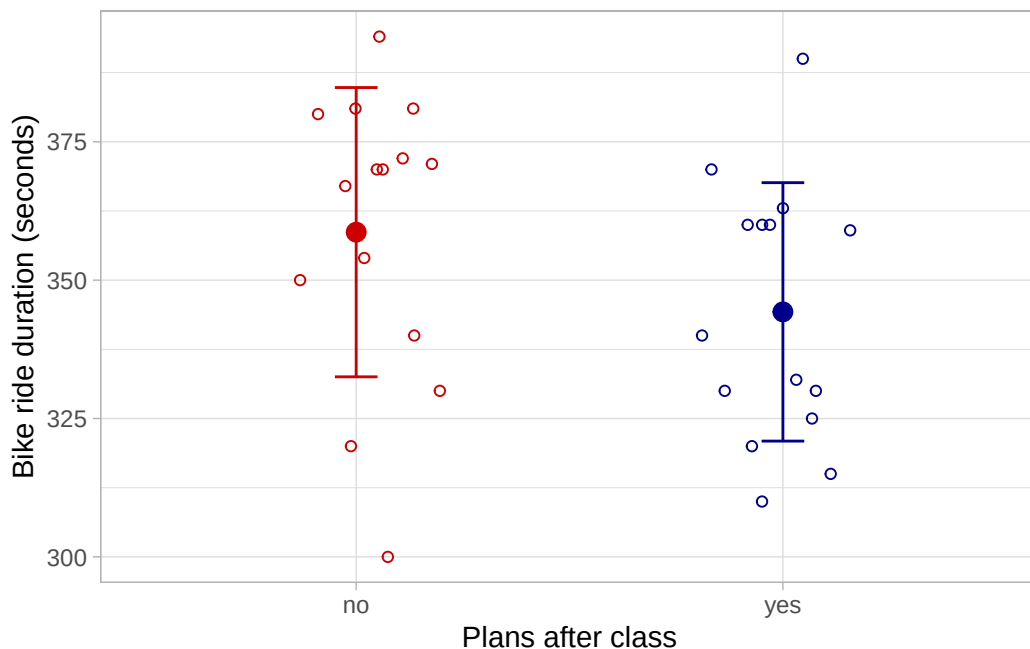
```
# create new object from bike_rides_clean
bike_rides_clean_summary <- bike_rides_clean |>
  group_by(plans_after_class) |>
```

```

summarize( # what I want in the summary
  mean = mean(duration_seconds),
  n = length(duration_seconds),
  sd = sd(duration_seconds)) |>
ungroup() # ungroup the data frame

# start with the bike_rides_clean data frame
ggplot(data = bike_rides_clean,
  # x-axis should be plans after class yes/no
  # y-axis should be duration in seconds
  # coloring by plans after class
  mapping = aes(x = plans_after_class,
    y = duration_seconds,
    color = plans_after_class)) +
# first layer: jitterplot
geom_jitter(height = 0, # no jitter in vertical direction
  width = 0.2, # small horizontal jitter
  shape = 21) + # open circle shape
# second layer: represent means for each group
stat_summary(geom = "point",
  fun = mean,
  size = 3) + # make points bigger
# third layer: bars to represent standard deviation
geom_errorbar(data = bike_rides_clean_summary, # use summary data frame
  mapping = aes(x = plans_after_class,
    y = mean, # assign axes based on summary frame
    ymin = mean - sd,
    ymax = mean + sd,
    width = 0.1)) + # make the bars narrower
# relabelling x- and y-axes
labs(x = "Plans after class",
  y = "Bike ride duration (seconds)") +
theme_light() + # custom theme
# custom colors
scale_color_manual(values = c("yes" = "darkblue",
  "no" = "red3")) +
theme(legend.position = "none") # taking out legend

```



- e. Figure 2. Bike ride duration in seconds dependent on whether I had plans after class. Open circles represent individual observations and large dots represent group means. The bars represent one standard deviation away from the mean in the positive and negative direction. Red corresponds with no plans after class, and blue corresponds with plans after class.
- f. We did not find evidence to suggest that whether I had plans after class has an effect on my mean bike ride durations (two sample t-test, $t(28) = 1.59$, $p=0.12$, $\alpha = 0.05$). On average, my bike ride durations in seconds when I didn't have plans after class ($n=15$, mean=359) were slightly longer than when I did have plans after class ($n=15$, mean=344). Having plans after class had a moderate effect on my bike ride durations (Cohen's $d = 0.58$), but the difference in means may have occurred by chance.